Who Pays the Bill?

Substrate Skepticism, Hidden Counterfactual Synergy, and an Information-Spectrum Tax

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Abstract

This note asks a deliberately narrow question. Suppose an observable system is specified only by controlled stochastic input–output laws. Without identifying the underlying ontology, what information resources must an exact realization carry once one specifies what is required to be jointly definite?

The first lesson is an accounting one. Chosen-path sampling, one-step counterfactual sampling, and a reusable counterfactual response tree are different realization contracts. Modern online randomness-recycling results show that chosen-path sampling can approach the Shannon entropy rate. A strict reusable-reservoir model, by contrast, exhibits an elementary spectrum-cancellation obstruction at exact Shannon equality.

The main purpose of this version is to remove the artificial reset independence and study finite counterfactual blocks directly. We define a *policy-safe block law*: a joint response table whose every action transversal has exactly the required product law. A transversal criterion shows that this is equivalent to exactness for every adaptive nonanticipative policy. For a concrete equal-entropy pair of three-outcome distributions we give an explicit two-step policy-safe law with hidden higher-order dependence and prove strict subadditivity of the minimum block entropy. Thus hidden correlations between mutually exclusive futures can genuinely reduce the counterfactual source cost.

They cannot, however, erase it completely. For two finite distributions P, Q, let μ_P, μ_Q denote the laws of their one-symbol surprisals $-\log_2 P(X)$ and $-\log_2 Q(Z)$. We derive the block lower bound

$$C_n^{\mathrm{ps}}(P,Q) \geq \frac{n}{2}\Big(H(P) + H(Q) + W_1(\mu_P, \mu_Q)\Big),$$

where W_1 is one-dimensional Wasserstein distance. Hence if $H(P) = H(Q) = h$ but the nonzero probability spectra are not permutations, then

$$C_\infty^{\mathrm{ps}}(P,Q) \geq h + \tfrac{1}{2} W_1(\mu_P, \mu_Q) > h.$$

For the explicit crash-test pair the current certified interval is

$$1.7284654687 \leq C_\infty^{\mathrm{ps}} \leq 1.9812960737 < C_1^{\mathrm{ps}} = 2.0539185856.$$

Thus counterfactual dependence permits netting, but a mismatch of self-information spectra leaves a positive asymptotic bill under this accounting standard.

We make no claim that the terminology or theorems are historically new. The construction lies close to minimum-entropy coupling, information-spectrum converses, sample-privacy synergy, all-or-nothing transforms, catalytic entropy, and finitary coding. The standing convention is that an equivalent or stronger result may already exist under different language unless a dedicated literature study establishes otherwise.

“The accountant discovered netting. The auditor discovered a lower bound.”

1 Priority convention and philosophical origin

1.1 Priority convention

The purpose of this note is to sharpen a resource question, not to manufacture a novelty claim. We therefore adopt the following standing rule.

> **Priority convention.** Assume that any clean statement derived here has probably been stated, proved, or embedded in a more general theory somewhere else. The word *derived* describes only the internal logic of this note. When a nearby precedent is known, it is cited; when no direct precedent has been located, we say only that we have not located one.

This caution is substantive. Minimum-entropy coupling is established information theory [2]; exact random-number generation and interval methods are classical [3, 4]; modern online samplers explicitly recycle unused randomness [5, 6, 7]. Ex-ante contingent plans versus local randomization have a long history in game theory [8, 9, 10]. Correlated and reversible catalysis show that auxiliary systems and correlations can remove obstructions that are visible in one-shot descriptions [13, 14, 15]. Joint MDPs separate per-action marginals from coupled counterfactual outcomes [11], while event-keyed common random numbers emphasize that cross-intervention identity itself is a modeling commitment [12].

The later sections add further neighbors: information-spectrum converses for minimum-entropy coupling [16], synergistic disclosure under perfect sample privacy [18], all-or-nothing transforms [19], marginally constrained dependent observations [17], and finitary isomorphisms of equal-entropy Bernoulli processes [20, 21, 22]. The exact combination of transversal constraints and the block objective used below was not located in this formulation in our search; this is not a priority assertion.

1.2 From substrate skepticism to resource accounting

A philosophical trigger was Pavel Arazim's public answer for the Czech outreach project *Zeptej se vědce* to the question "What if life is a dream and one wakes up after death?" [1]. The connection here is motivational only; it does not imply endorsement of the mathematics below.

Instead of asking

"What is the substrate?"

we ask the weaker operational question

$$\boxed{\text{If an exact substrate exists under a stated contract, what resources must it carry?}}$$

We call this stance *substrate skepticism*. An implementation lower bound is not an ontological identification. Conversely, refusing ontology does not make exact realization resource-free.

2 The accounting-standard principle

A recurrent ambiguity in discussions of randomness, simulation, and "fundamental reality" is that different realization contracts are silently exchanged. We distinguish three levels.

C1. Chosen-path sampling. At time t, the action A_t is already known. Only the requested output $Y_t \sim P_{A_t}$ must be generated.

C2. One-step counterfactual table. Before the action is selected, one latent draw determines

$$F_t = (F_t(a))_{a \in \mathcal{A}}, \qquad F_t(a) \sim P_a.$$

Only one coordinate is observed, but all coordinates are jointly defined.

C3. Reusable counterfactual substrate. Counterfactual information survives across time and must remain valid under future adaptive action choices. This is the natural contract for a single coherent response tree.

The first principle is therefore:

$$\boxed{\text{The bill is meaningful only after the accounting standard is fixed.}}$$

3 Controlled channels and the one-shot bill

Let $\mathcal{A}$ and $\mathcal{Y}$ be finite. A memoryless controlled channel is a family

$$W = \{P_a : a \in \mathcal{A}\}, \qquad P_a \in \Delta(\mathcal{Y}),$$

satisfying, for every adaptive action process,

$$\Pr(Y_t = y \mid A_{1:t}, Y_{1:t-1}) = P_{A_t}(y). \tag{3.1}$$

The controller may use any deterministic nonanticipative policy

$$A_t = \pi_t(A_{1:t-1}, Y_{1:t-1}),$$

and randomized policies can be included by adding private policy randomness independent of the channel.

Define

$$h_{\max} := \max_{a \in \mathcal{A}} H(P_a). \tag{3.2}$$

Along the policy that always chooses an entropy-maximizing action, any exact implementation must supply at least $h_{\max}$ bits per step in the appropriate asymptotic sense.

Under C2, a single latent table $F = (F(a))_{a \in \mathcal{A}}$ has prescribed coordinate marginals. Its minimum entropy is exactly the multi-marginal minimum-entropy coupling

$$C_1(W) := \min\{H(F) : \mathrm{Law}(F(a)) = P_a \ \forall a \in \mathcal{A}\} = \mathrm{MEC}(P_a : a \in \mathcal{A}). \tag{3.3}$$

The one-shot surcharge is

$$r_1(W) := C_1(W) - h_{\max} \geq 0. \tag{3.4}$$

The central mistake to avoid is multiplying r_1 by time without proving that the realization contract requires independent one-shot tables.

4 Chosen-path sampling nearly reaches Shannon

Under C1, modern online sampling already provides a strong negative result against a naive per-step counterfactual tax. Draper and Saad construct randomness-recycling samplers with amortized cost arbitrarily close to the Shannon rate using logarithmic persistent state in the entropy slack [5]. Draper, Harris, and Saad extend the picture to real-valued probabilities and obtain, for adaptive target laws $F_1, F_2, \ldots$, expected coin consumption

$$\mathbb{E}[T_n] \leq \mathbb{E}\left[\sum_{t=1}^{n} H(F_t)\right] + O(\log n), \tag{4.1}$$

with logarithmic persistent space in their model [6]. Complementary lower bounds show that uniformly approaching the Shannon rate forces persistent space to grow at least logarithmically in the inverse entropy slack over broad classes [7].

Thus adaptive choice alone does not force a positive entropy-rate surcharge. The unresolved resource is the cost of keeping mutually exclusive interventions jointly meaningful.

5 Warm-up: a strict reusable-reservoir obstruction

Before allowing hidden correlations, it is useful to record the elementary obstruction that motivated the later block problem.

Definition 5.1 (Strong reset recycler)**.** Fix a finite reservoir law λ on $\mathcal{S}$ and a finite innovation law ν on $\mathcal{U}$. A strong reset recycler consists of deterministic maps

$$\Phi_a : \mathcal{S} \times \mathcal{U} \to \mathcal{Y} \times \mathcal{S}, \qquad a \in \mathcal{A},$$

such that for independent $S \sim \lambda$ and $U \sim \nu$,

$$\Phi_a(S, U) = (Y, S') \sim P_a \otimes \lambda \qquad \forall a \in \mathcal{A}. \tag{5.1}$$

The reservoir is therefore returned with its original marginal law and is independent of the newly exposed output.

Proposition 5.2 (Entropy balance)**.** *Every strong reset recycler satisfies*

$$H(\nu) \geq \max_{a \in \mathcal{A}} H(P_a). \tag{5.2}$$

Proof. For each a, (Y, S') is a deterministic function of (S, U), hence

$$H(P_a) + H(\lambda) = H(Y, S') \leq H(S, U) = H(\lambda) + H(\nu).$$

$\square$

Theorem 5.3 (Exact floor forces spectral equivalence)**.** *Let P, Q be finite probability vectors with $H(P) = H(Q) = h$. If one finite strong reset recycler serves both actions and satisfies $H(\nu) = h$, then P and Q have the same multiset of nonzero atom probabilities, up to zero padding.*

Proof. For action P, entropy balance is tight. Since (Y, S') is a deterministic function of (S, U),

$$H(S, U \mid Y, S') = 0.$$

Thus the map is invertible almost surely on positive-probability atoms and

$$\lambda \otimes \nu \equiv_{\text{perm}} P \otimes \lambda.$$

The same argument for Q gives

$$P \otimes \lambda \equiv_{\text{perm}} Q \otimes \lambda. \tag{5.3}$$

For every integer $r \geq 2$, equality of product atom multisets yields

$$\left(\sum_i p_i^r\right)\left(\sum_j \lambda_j^r\right) = \left(\sum_i q_i^r\right)\left(\sum_j \lambda_j^r\right).$$

The second factor is positive, hence all power sums of P and Q agree. After zero padding, Newton's identities determine the multiset of atoms. □

This theorem is a crash-test for one accounting standard, not a universal rate theorem. Correlated catalysis gives ample reason not to assume the reset independence without proof [13, 14].

6 A concrete equal-entropy crash test

Consider

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \qquad Q = (q, q, 1-2q), \tag{6.1}$$

where $q \in (1/3, 1/2)$ is the unique solution of

$$-2q \log_2 q - (1-2q)\log_2(1-2q) = \frac{3}{2}. \tag{6.2}$$

Numerically,

$$q = 0.41003242818198557608\ldots, \qquad H(P) = H(Q) = 1.5\,\text{bits}. \tag{6.3}$$

The spectra are not permutations.

A direct enumeration of the vertices of the 3×3 transportation polytope gives a one-shot minimum-entropy coupling with nonzero atom probabilities

$$\frac{1}{2} - q, \quad q, \quad 2q - \frac{3}{4}, \quad 1 - 2q, \quad \frac{1}{4}, \tag{6.4}$$

and therefore

$$C_1(P, Q) = \mathrm{MEC}(P, Q) = 2.0539185856072597\ldots\ \text{bits}. \tag{6.5}$$

The one-shot surcharge above the observable Shannon floor is about 0.5539186 bits.

The strict reset model cannot attain 1.5 bits at any finite reservoir size by Theorem 5.3. The important question is what happens after the reset independence is removed.

7 Policy-safe counterfactual blocks

For a block of length n, let

$$F = (F_t(a))_{1 \leq t \leq n,\ a \in \mathcal{A}} \tag{7.1}$$

be one jointly distributed counterfactual response table.

Definition 7.1 (Policy-safe block law)**.** A law Γ of F is *policy-safe* if for every deterministic action sequence $a^n = (a_1, \ldots, a_n)$,

$$\mathrm{Law}_\Gamma(F_1(a_1), \ldots, F_n(a_n)) = \bigotimes_{t=1}^{n} P_{a_t}. \tag{7.2}$$

The definition mentions fixed action strings only, yet it already controls adaptive policies.

Proposition 7.2 (Transversal criterion)**.** *A block law is policy-safe if and only if it realizes the exact memoryless channel for every deterministic nonanticipative policy. The same holds for randomized policies with independent private randomization.*

Proof. Necessity follows by considering policies that ignore the history. For sufficiency, fix a deterministic policy and an output string y^n. Along this branch the policy determines the action string

$$a_t = a_t(y_{1:t-1}).$$

Hence

$$\{Y^n = y^n\} = \bigcap_{t=1}^{n} \{F_t(a_t(y_{1:t-1})) = y_t\}.$$

Policy-safety gives

$$\Pr(Y^n = y^n) = \prod_{t=1}^{n} P_{a_t(y_{1:t-1})}(y_t),$$

which is exactly the channel law under that policy. Randomized policies follow by conditioning on their independent private seed. □

Define the minimum policy-safe block entropy

$$C_n^{\mathrm{ps}}(W) := \min\{H(F) : \mathrm{Law}(F) \text{ is policy-safe}\}. \tag{7.3}$$

The feasible set is a nonempty compact polytope, so the minimum exists.

Proposition 7.3 (Elementary bounds and subadditivity)**.** *For every n,*

$$nh_{\max} \le C_n^{\mathrm{ps}}(W) \le nC_1(W), \tag{7.4}$$

and

$$C_{m+n}^{\mathrm{ps}}(W) \le C_m^{\mathrm{ps}}(W) + C_n^{\mathrm{ps}}(W). \tag{7.5}$$

Consequently

$$C_\infty^{\mathrm{ps}}(W) := \lim_{n\to\infty} \frac{C_n^{\mathrm{ps}}(W)}{n} = \inf_{n\ge 1} \frac{C_n^{\mathrm{ps}}(W)}{n}. \tag{7.6}$$

Proof. Select an entropy-maximizing action at every time. Its transversal has entropy $nh_{\max}$ and is a deterministic projection of F. Independent optimal one-step couplings give the upper bound. Independent concatenation of optimal m- and n-blocks preserves every transversal product law and gives subadditivity. Fekete's lemma yields the limit. □

For two actions, write

$$X_t := F_t(P), \qquad Z_t := F_t(Q).$$

Then policy-safety is equivalently the family of constraints

$$(X_S, Z_{S^c}) \sim P^{\otimes S} \otimes Q^{\otimes S^c} \qquad \forall S \subseteq [n], \tag{7.7}$$

where at each time exactly one of the two counterfactual coordinates is selected. This form makes clear that the object is stronger than an ordinary coupling of two Bernoulli shifts: every mixed transversal must itself be an independent product process.

8 Hidden synergy is an operational resource

The policy-safe constraints do not force the full tables at different times to be independent. In fact, the slack can be used.

For one time step let

$$r = (F(P), F(Q)) \in \{0, 1, 2\}^2$$

and order the nine states as

$$\mathcal{R} = (00, 01, 02, 10, 11, 12, 20, 21, 22). \tag{8.1}$$

Appendix A gives an explicit 9×9 matrix G whose rows index $F_1 \in \mathcal{R}$ and columns index $F_2 \in \mathcal{R}$.

Proposition 8.1 (Explicit two-step hidden-synergy witness)**.** *At the value of q in (6.2), the matrix G in Appendix A is a probability law with 25 positive atoms. For every $a, b \in \{P, Q\}$,*

$$(F_1(a), F_2(b)) \sim P_a \otimes P_b. \tag{8.2}$$

Hence G is policy-safe for two steps.

Proof. The entries sum symbolically to one. At q in (6.2), all displayed nonzero entries are positive; the smallest is approximately

$$6.65947662069 \times 10^{-5}.$$

Direct symbolic summation of the four 3×3 projections gives exactly

$$P \otimes P, \qquad P \otimes Q, \qquad Q \otimes P, \qquad Q \otimes Q.$$

Proposition 7.2 then gives adaptive exactness. $\square$

Let $F_t = (F_t(P), F_t(Q))$. For this witness,

$$\begin{aligned} H(F_1) &= 2.7473113605864043\ldots, \\ H(F_2) &= 2.4591672061227527\ldots, \\ H(F_1, F_2) = H(G) &= 3.9625921473435652\ldots, \end{aligned} \tag{8.3}$$

and therefore

$$I(F_1; F_2) = 1.2438864193655918\ldots \text{ bits}. \tag{8.4}$$

More specifically,

$$I(F_1(P); F_2) = 0.1944768392850001\ldots \text{ bits}, \tag{8.5}$$

while

$$I(F_1(P); F_2(P)) = I(F_1(P); F_2(Q)) = 0. \tag{8.6}$$

Likewise,

$$I(F_1(Q); F_2) = 0.1813243457087152\ldots \text{ bits}, \tag{8.7}$$

but each individual future coordinate is independent of $F_1(Q)$.

Thus an observed past can carry information about the *joint* future counterfactual table while revealing nothing about any future coordinate that the controller can individually select. In this note, this is the precise meaning of *hidden synergy*:

$$\boxed{I(Y_{\text{past}}; F_{\text{future}}) > 0, \qquad I(Y_{\text{past}}; F_{\text{future}}(a)) = 0 \quad \forall a.} \tag{8.8}$$

Closely related forms of "jointly informative but individually private" dependence occur in synergistic disclosure and all-or-nothing-transform literature [18, 19]. The present constraints and optimization objective are different; the terminology is not claimed as new.

The construction has an immediate operational consequence.

Corollary 8.2 (Strict block subadditivity for the crash test)**.** *For the pair in Section 6,*

$$C_2^{\mathrm{ps}}(P,Q) \le H(G) < 2C_1(P,Q). \tag{8.9}$$

Numerically,

$$2C_1 - H(G) = 0.1452450238709542\ldots \text{ bits.} \tag{8.10}$$

Hence

$$C_\infty^{\mathrm{ps}}(P,Q) \le \frac{H(G)}{2} = 1.9812960736717826\ldots \text{ bits.} \tag{8.11}$$

This proves that hidden cross-counterfactual dependence is not merely possible: it strictly lowers the block source entropy relative to independent one-step minimum-entropy couplings.

9 The information-spectrum transport bound

Hidden synergy reduces the bill, but it does not have unlimited freedom. The key observation is pointwise and does not assume a reset state, Markov structure, finite memory, or independence between time slices.

For a finite probability vector $P = (p_i)$ define its one-symbol surprisal random variable

$$\imath_P(X) := -\log_2 p_X, \qquad X \sim P, \tag{9.1}$$

and its *surprisal law*

$$\mu_P := \sum_i p_i \, \delta_{-\log_2 p_i}. \tag{9.2}$$

Let W_1 denote the 1-Wasserstein distance on $\mathbb{R}$ with cost $|u-v|$.

Lemma 9.1 (Surprisal transport identity)**.** *For finite P, Q,*

$$\inf_{\gamma \in \Pi(P,Q)} \mathbb{E}_\gamma \big|\imath_P(X) - \imath_Q(Z)\big| = W_1(\mu_P, \mu_Q), \tag{9.3}$$

where $\Pi(P,Q)$ is the set of couplings with marginals P, Q.

Proof. Every atom-level coupling γ pushes forward under $(i,j) \mapsto (-\log_2 p_i, -\log_2 q_j)$ to a coupling of μ_P and μ_Q, giving the "$\ge$" direction.

Conversely, let η couple μ_P and μ_Q. For a surprisal level u, all atoms in

$$A_u := \{i : -\log_2 p_i = u\}$$

have the same mass 2^{-u}; similarly define B_v. If $m_u = |A_u|$ and $n_v = |B_v|$, distribute $\eta(u,v)$ uniformly over $A_u \times B_v$:

$$\gamma(i,j) = \frac{\eta(u,v)}{m_u n_v} \qquad (i \in A_u, j \in B_v).$$

The row mass of each $i \in A_u$ is

$$\frac{1}{m_u} \sum_v \eta(u,v) = \frac{\mu_P(u)}{m_u} = 2^{-u} = p_i,$$

and similarly for columns. Thus every surprisal-level transport plan lifts to an atom-level coupling without changing cost. $\square$

Theorem 9.2 (Policy-safe information-spectrum lower bound)**.** *Let P, Q be finite distributions and let $F = (X_1, Z_1, \dots, X_n, Z_n)$ be any policy-safe two-action block law. Then*

$$H(F) \geq \frac{n}{2}\Big(H(P) + H(Q) + W_1(\mu_P, \mu_Q)\Big). \tag{9.4}$$

Consequently,

$$\boxed{C_\infty^{\mathrm{ps}}(P, Q) \geq \frac{1}{2}\Big(H(P) + H(Q) + W_1(\mu_P, \mu_Q)\Big).} \tag{9.5}$$

Proof. Fix a positive-probability full-table atom

$$f = (x_1, z_1, \dots, x_n, z_n)$$

with probability $\Gamma(f)$. For each subset $S \subseteq [n]$, this full atom lies inside the transversal event

$$X_S = x_S, \qquad Z_{S^c} = z_{S^c}.$$

By policy-safety,

$$\Gamma(f) \leq \prod_{t \in S} P(x_t) \prod_{t \notin S} Q(z_t). \tag{9.6}$$

Minimizing over S separates coordinatewise:

$$\Gamma(f) \leq \prod_{t=1}^{n} \min\{P(x_t), Q(z_t)\}. \tag{9.7}$$

Therefore

$$-\log_2 \Gamma(f) \geq \sum_{t=1}^{n} \max\{\imath_P(x_t), \imath_Q(z_t)\}. \tag{9.8}$$

Average over F. The one-time pair (X_t, Z_t) has some coupling $\gamma_t \in \Pi(P, Q)$, because policy-safety fixes each coordinate marginal. Hence

$$\begin{aligned} H(F) &\geq \sum_{t=1}^{n} \mathbb{E}_{\gamma_t} \max\{\imath_P(X), \imath_Q(Z)\} \\ &= \frac{1}{2}\sum_{t=1}^{n} \left(H(P) + H(Q) + \mathbb{E}_{\gamma_t}|\imath_P(X) - \imath_Q(Z)|\right) \\ &\geq \frac{n}{2}\Big(H(P) + H(Q) + W_1(\mu_P, \mu_Q)\Big), \end{aligned}$$

where the last step uses Lemma 9.1. Taking the infimum over policy-safe laws and then the asymptotic limit proves (9.5). ☐

Remark 9.3 (Relation to information-spectrum converses)*.* The proof is an information-spectrum argument tailored to the simultaneous transversal constraints. General information-spectrum converses for minimum-entropy couplings and functional representations are already known [16]. We have not established whether Theorem 9.2 is a direct corollary of a previously published converse after a suitable reformulation; no novelty claim is made.

The theorem immediately sharpens in the equal-entropy case.

Corollary 9.4 (Equal entropy, unequal spectra)**.** *If*

$$H(P) = H(Q) = h,$$

then

$$C_\infty^{\mathrm{ps}}(P, Q) \geq h + \frac{1}{2} W_1(\mu_P, \mu_Q). \tag{9.9}$$

Moreover,

$$W_1(\mu_P, \mu_Q) = 0$$

if and only if P and Q have the same multiset of nonzero atom probabilities. Hence any equal-entropy non-permutation pair satisfies

$$\boxed{C_\infty^{\mathrm{ps}}(P, Q) > h.} \tag{9.10}$$

Proof. Only the characterization of equality requires comment. If the probability spectra agree, the surprisal laws agree. Conversely, a mass value $r > 0$ corresponds to surprisal $-\log_2 r$. If that level has multiplicity m_r, the mass of μ_P at that surprisal is $m_r r$. Thus μ_P determines m_r for every nonzero atom size r. Equality of the surprisal laws therefore determines the probability spectrum up to permutation and zero padding. □

The strong-reset theorem and the block theorem therefore detect the same underlying mismatch in two different ways. Under exact reset equality, tensor-product cancellation preserves the full atom spectrum. Under arbitrary policy-safe hidden dependence, the surviving obstruction is a positive transport distance between the one-step self-information spectra.

10 The crash test after the lower bound

For $P = (1/2, 1/4, 1/4)$,

$$\mu_P = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2. \tag{10.1}$$

For $Q = (q, q, 1 - 2q)$,

$$\mu_Q = 2q\,\delta_{-\log_2 q} + (1 - 2q)\,\delta_{-\log_2(1-2q)}. \tag{10.2}$$

Numerically,

$$\begin{aligned} -\log_2 q &= 1.2861900824071589\ldots, \\ -\log_2(1 - 2q) &= 2.4744511039746512\ldots. \end{aligned} \tag{10.3}$$

The one-dimensional optimal transport distance is

$$W_1(\mu_P, \mu_Q) = 0.45693093749106154\ldots, \tag{10.4}$$

and Theorem 9.2 gives

$$C_\infty^{\mathrm{ps}}(P, Q) \geq 1.5 + \frac{1}{2}W_1(\mu_P, \mu_Q) = 1.7284654687455308\ldots. \tag{10.5}$$

Combining this with the explicit two-step witness gives the current certified interval

$$\boxed{1.7284654687455308 \leq C_\infty^{\mathrm{ps}}(P, Q) \leq 1.9812960736717826.} \tag{10.6}$$

Both inequalities are strict relative to the endpoints that motivated the problem:

$$1.5 < C_\infty^{\mathrm{ps}}(P, Q) < C_1(P, Q) = 2.0539185856072597\ldots. \tag{10.7}$$

Thus the hidden-synergy attack has a two-sided answer:

$$\boxed{\text{hidden correlations reduce the bill, but cannot erase it.}}$$

11 From block entropy to fresh innovation: an explicit bridge

The block quantity C_n^{ps} is the entropy of a source that fixes a finite counterfactual response table. It should not be silently identified with every possible notion of dynamic fresh-randomness consumption. A bridge requires an implementation contract.

One useful contract is the following. Suppose that for every horizon n a policy-safe table $F^{(n)}$ is a deterministic function of an initial reservoir S_0 and an external innovation string $U_{1:n}$:

$$F^{(n)} = \phi_n(S_0, U_{1:n}), \tag{11.1}$$

with S_0 independent of the innovation source. No Markov assumption is needed.

Proposition 11.1 (Finite-preload innovation lower bound)**.** *Under* (11.1)*,*

$$H(U_{1:n}) \geq C_n^{\mathrm{ps}}(W) - H(S_0). \tag{11.2}$$

For two actions P, Q, if $H(S_0) = o(n)$, then

$$\liminf_{n\to\infty} \frac{1}{n} H(U_{1:n}) \geq \frac{1}{2}\Big(H(P) + H(Q) + W_1(\mu_P, \mu_Q)\Big). \tag{11.3}$$

Proof. Since $F^{(n)}$ is deterministic given $(S_0, U_{1:n})$,

$$C_n^{\mathrm{ps}}(W) \leq H(F^{(n)}) \leq H(S_0, U_{1:n}) \leq H(S_0) + H(U_{1:n}).$$

Apply Theorem 9.2 and divide by n. □

This proposition is intentionally conditional. An implementation that stores an extensive amount of preloaded information, treats infinite precision as free, or uses a different branch-indexed source model changes the accounting standard. The mathematical lower bound does not license an ontological conclusion by itself.

12 Neighboring theories and why they do not collapse the problem

Several known theories illuminate pieces of the construction.

Minimum-entropy coupling and information spectrum. The one-shot quantity C_1 is classical MEC [2]. Shkel and Yadav develop general information-spectrum converses and improved lower bounds for MEC and functional representations [16]. Theorem 9.2 should be read in that neighborhood.

Dependent observations under fixed marginals. Asoodeh and Chen show that allowing dependence among marginally constrained observations can qualitatively change finite-sample entropy problems and can even enable exact finite-sample recovery in settings where independent observations cannot [17]. Their feasible set and objective are different from policy-safe counterfactual tables, but the moral is highly relevant: dependence can be an operational resource rather than a nuisance.

Synergistic disclosure and AONTs. Perfect sample privacy permits a disclosed variable to carry collective information while revealing no information about individual samples [18]. All-or-nothing transforms study exact subset-independence properties and their dependence on the input law [19]. These are close conceptual neighbors of the hidden-synergy witness, though neither imposes the full family of mixed action transversals in (7.7).

Finitary Bernoulli coding. Equal-entropy finite-alphabet Bernoulli schemes are finitarily isomorphic [20]. Coding-radius moments carry additional information: Harvey and Peres identified informational variance as an obstruction under a finite square-root moment condition [21], while Gabor obtained near-optimal positive results below the 1/2 moment threshold [22]. These results warn that a one-step spectrum mismatch can sometimes be rearranged across time if the coding window is allowed to grow. They do not settle the present problem because policy-safety requires every mixed transversal, not only a recoding from one entire process into another.

Counterfactual dynamics and stable randomness identity. Joint MDPs explicitly attach a coupling to multi-action one-step outcomes [11]. Event-keyed hashing for common random numbers shows that even the phrase "the same randomness across interventions" requires a stable notion of event identity [12]. A full response tree therefore contains a causal indexing commitment beyond ordinary marginal correctness.

13 Precision, memory, and counterfactual breadth

A scalar entropy rate is still not the whole bill.

First, exact real-valued state can hide unbounded discrete precision. If

$$U \sim \mathrm{Unif}[0,1],$$

then, outside the null set of dyadic ambiguities, its binary expansion contains an infinite fair-bit tape. Treating one exact real number as a finite-cost state can therefore prepay arbitrarily much future randomness. A resource theory that allows exact real computation must price precision, description complexity, or oracle access; differential entropy alone is not enough.

Second, near-Shannon online sampling already exhibits a memory–innovation tradeoff [5, 6, 7]. The policy-safe block problem adds another deformation: hidden counterfactual dependence can lower source entropy, but Theorem 9.2 leaves a positive rate whenever surprisal spectra differ.

Third, the breadth of the intervention family matters. A finite collection of policies can share only finitely much response-tree structure. The class of all nonanticipative policies up to horizon n continually regenerates branching constraints at every depth. A useful resource geometry therefore has at least the schematic coordinates

$$(M, R, \Pi, B), \tag{13.1}$$

where M is persistent latent memory, R fresh innovation, Π precision or description complexity, and B counterfactual breadth.

The original metaphor of one bill is therefore deliberately incomplete. There are exchange rates between memory, innovation, precision, and the amount of counterfactual structure required to remain jointly definite.

14 What is proved, what is not, and what remains open

The technical conclusions can be stated compactly.

Proved under the policy-safe block contract

- Adaptive exactness is equivalent to the transversal constraints of Proposition 7.2.
- C_n^{ps} is subadditive and has a well-defined asymptotic rate.

- For the explicit equal-entropy pair, hidden synergy exists and strictly reduces block source entropy: $C_2^{\mathrm{ps}} < 2C_1$.
- Every two-action policy-safe block obeys the information-spectrum transport lower bound (9.4).
- Equal Shannon entropy is not an asymptotically complete invariant: non-permutation spectra force $C_\infty^{\mathrm{ps}} > h$.
- For the crash test, $C_\infty^{\mathrm{ps}} \in [1.7284654687, 1.9812960737]$.

Not proved

- The explicit matrix G is not claimed to minimize C_2^{ps}; it is a feasible witness.
- The exact value of C_∞^{ps} for the crash-test pair is unknown.
- No universal theorem is claimed for arbitrary notions of dynamic reusable reservoirs unless they induce the source model specified in Section 11 or another explicit bridge is proved.
- No entropy inequality here identifies the physical substrate, disproves simulation, or decides whether observed randomness is ontic or epistemic.
- No historical novelty is claimed for the terminology or the derived inequalities.

Open Problem 14.1 (Exact asymptotic policy-safe rate)*.* For the pair in Section 6, determine

$$C_\infty^{\mathrm{ps}}(P, Q).$$

Can the current upper bound 1.9812960737 be pushed down to the transport lower bound 1.7284654687, or is there a stronger invariant generated by the simultaneous transversal constraints?

Open Problem 14.2 (Multi-action information-spectrum geometry)*.* For $|\mathcal{A}| > 2$, find the sharp analogue of Theorem 9.2. The pointwise atom bound becomes a minimum over actions at each time, suggesting a multi-marginal optimal-transport problem over surprisal vectors rather than the one-dimensional W_1 term.

Open Problem 14.3 (Dynamic equivalence)*.* Characterize the classes of reusable-reservoir machines for which asymptotic fresh-innovation rate is exactly captured by C_∞^{ps} up to sublinear preload. Identify the additional resource term when branch-indexed innovations, unbounded precision, or extensive initial memory are permitted.

Open Problem 14.4 (Optimal hidden synergy)*.* Characterize the maximal entropy saving obtainable from dependence that is invisible on every admissible realized transversal. In particular, determine when strict subadditivity $C_n^{\mathrm{ps}} < nC_1$ first appears and how it scales with n.

15 Substrate skepticism after the mathematics

The mathematical lesson remains narrower than the metaphor.

A chosen-path simulator can recycle randomness almost down to Shannon. A strict independent-reset reservoir can fail at exact Shannon equality because probability spectra do not cancel. Once hidden correlations are allowed, that one-shot obstruction weakens: the explicit two-step witness proves that mutually exclusive futures can share information and reduce the block bill. But the policy-safe transport theorem leaves a positive asymptotic surcharge whenever the self-information spectra differ.

This does *not* tell us what lies underneath. A fundamental stochastic process, a simulator, a deeper deterministic system with a sufficiently rich initial condition, or another mathematically equivalent representation may agree on all observations while paying under different resource coordinates. Precision can substitute for stored randomness; extensive preload can substitute for innovation; refusing full counterfactual definiteness can remove response-tree obligations.

The strongest substrate-skeptical statement supported here is therefore:

> We do not infer what reality is made of. We specify what an exact realization is required to preserve, then ask for the cheapest admissible implementation. Under a policy-safe full-counterfactual block contract, equal Shannon entropy is not enough: a mismatch of self-information spectra leaves a positive asymptotic source cost even after arbitrary hidden cross-counterfactual correlations are allowed.

The accountant may consolidate the invoice. He may refinance it across mutually exclusive futures. Under this accounting standard, he cannot write it off completely.

16 Conclusion

The attack changed the target several times. The one-shot MEC surcharge is real but is not automatically a per-step tax. Chosen-path randomness can be recycled close to the Shannon rate. Strong independent reset creates a rigid spectrum-cancellation obstruction, but that reset condition is unnecessarily strong. Removing it reveals hidden synergy: information can live only in relations among mutually exclusive futures and can strictly lower counterfactual block entropy.

The resulting asymptotic problem nevertheless has a nonzero floor. The pointwise fact that a full response-table atom must fit inside *every* observable transversal yields an information-spectrum transport converse. For two equal-entropy distributions, the positive quantity

$$\frac{1}{2}W_1(\mu_P, \mu_Q)$$

is a counterfactual source-rate surcharge whenever the probability spectra differ.

For the concrete three-outcome pair,

$$1.5 < 1.7284654687 \le C_\infty^{\text{ps}} \le 1.9812960737 < 2.0539185856 = C_1.$$

The one-shot bill was too large. The Shannon bill was too small. Hidden correlations pay part of the difference.

The ontological question remains deliberately unanswered:

$$\boxed{\text{Who pays?}}$$

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The author is not writing from an academic position in information theory, probability, causal inference, or mathematical physics. The investigation is amateur in the literal sense: it is pursued because the questions are interesting. The standing assumption remains that any elegant theorem encountered along the way probably has a prior owner whom we have not yet found.

A The explicit two-step witness

With rows and columns ordered by

$$\mathcal{R} = (00, 01, 02, 10, 11, 12, 20, 21, 22),$$

the two-step law used in Proposition 8.1 is

$$G = \begin{pmatrix} 0 & q^2 & 0 & q/4 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{(3q-1)(4q-1)}{4} & -\frac{q(16q-7)}{4} & 0 & 0 & -\frac{2q-1}{8} & \frac{(4q-1)^2}{4} & -\frac{16q^2-9q+1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{16q^2-9q+1}{4} & 0 & \frac{(4q-1)(8q-3)}{8} \\ -\frac{8q^2+4q-3}{8} & 0 & 0 & 0 & 0 & 0 & \frac{32q^2+4q-7}{16} & 0 & -\frac{16q^2-3}{8} \\ -\frac{(2q-1)(4q-1)}{8} & q/4 & \frac{16q^2-4q-1}{8} & -\frac{(2q-1)(8q-1)}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(4q-1)(8q-3)}{16} & 0 & 0 & -\frac{2q-1}{8} & 0 & 0 \\ 0 & 0 & \frac{8q-3}{8} & 0 & 0 & 0 & \frac{1}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{q(8q-3)}{4} & 0 & 0 & 0 & 0 \\ \frac{(2q-1)^2}{2} & -q(2q-1) & 0 & -\frac{(4q-3)(8q-3)}{16} & 0 & -\frac{12q-5}{8} & 0 & 0 & 0 \end{pmatrix}. \tag{A.1}$$

The four transversal projection identities are polynomial identities in q and do not use the entropy equation (6.2); that equation is needed only to select the equal-entropy crash-test value. At that value all 25 displayed nonzero atoms are positive.

B Reproducibility checks

A companion script, `wpb_v2_verify.py`, performs the following checks at high precision:

1. solves (6.2) for q;
2. verifies symbolically that $\sum G_{rs} = 1$;
3. verifies symbolically all four 3×3 transversal projections in (8.2);
4. checks positivity and counts the 25 nonzero atoms;
5. evaluates $H(G)$ and the hidden-synergy mutual informations;
6. evaluates the one-shot MEC witness, strict subadditivity margin, $W_1(\mu_P, \mu_Q)$, and the lower bound (10.5).

The numerical values quoted in this manuscript are regenerated by that script rather than entered independently.

C Literature map rather than novelty claim

Topic	Status used here	Representative neighbor
Minimum-entropy coupling	Known	Kovačević–Stanojević–Šenk [2]
Information-spectrum MEC converses	Known	Shkel–Yadav [16]
Online randomness recycling	Known	Draper–Saad [5]
Near-Shannon real-probability sampling	Known	Draper–Harris–Saad [6]
Space lower bounds near Shannon	Known	Draper–Saad [7]
Joint one-step counterfactual coupling	Known	Kaya–Ghasemi–Hashemi [11]
Synergistic disclosure / sample privacy	Known	Rassouli–Rosas–Gündüz [18]
Subset-independence via AONTs	Known	Esfahani–Stinson [19]
Dependent marginally constrained observations	Known	Asoodeh–Chen [17]
Equal-entropy finitary Bernoulli coding	Known	Keane–Smorodinsky; Harvey–Peres; Gabor [20, 21, 22]
Policy-safe block entropy C_n^{ps}	Not located in this formulation	No direct source located in the present search
Transport bound (9.4) under all transversals	Not located in this formulation	Compare information-spectrum MEC converses [16]
Exact value of the crash-test C_∞^{ps}	Open here	Current interval (10.6)

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